

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2



**ST. ANDREW'S JUNIOR COLLEGE
2018 JC2 PRELIMS**

H2 BIOLOGY

9744/1

Paper 1: Multiple Choice (MARK SCHEME)

Tuesday

18th Sept 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **30** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

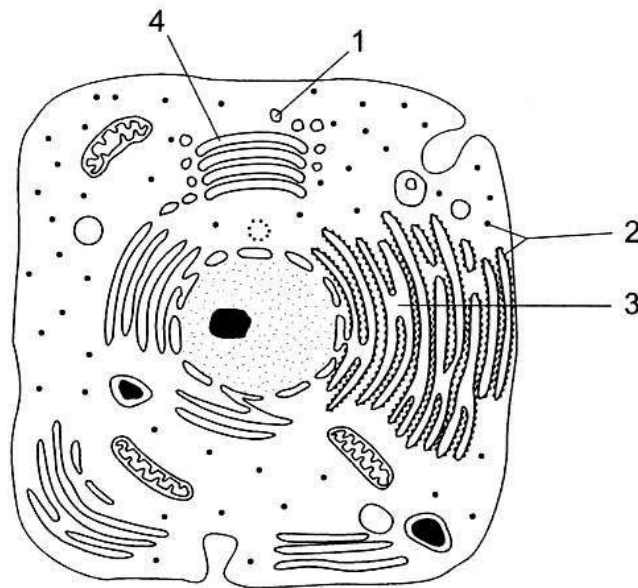
Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit both question paper and multiple choice answer sheet.

This document consists of **21** printed pages.

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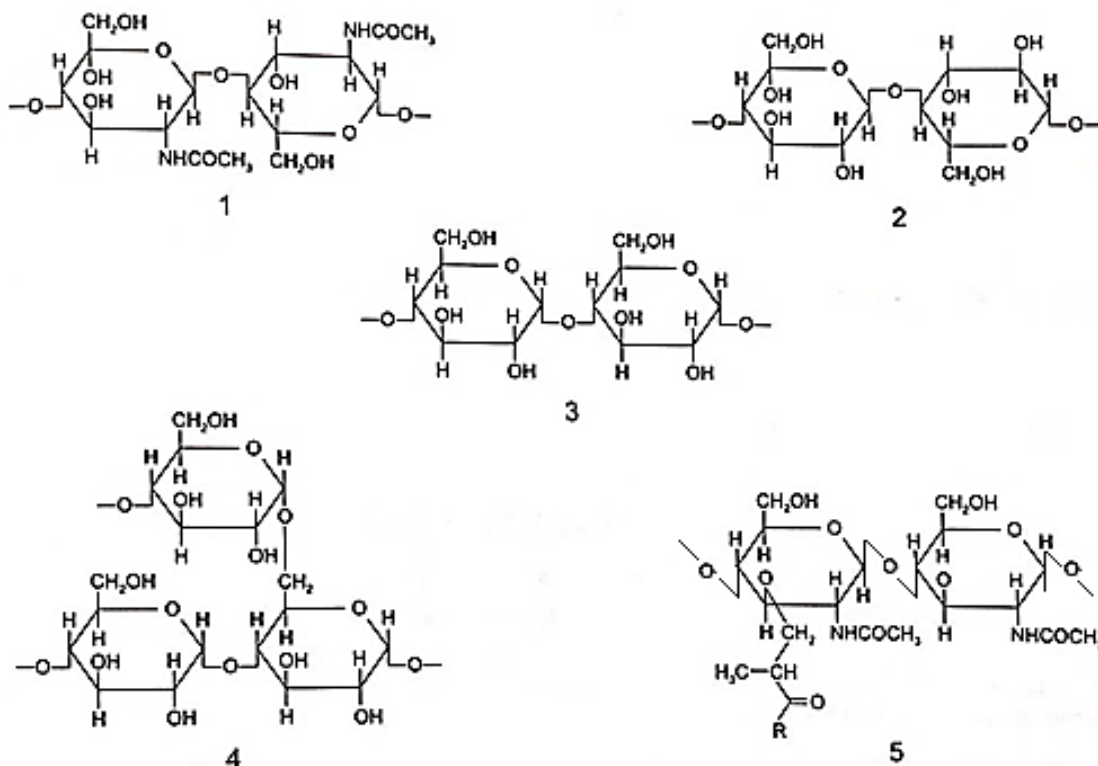
1 The figure below shows the structure of an animal cell.



Which of the following correctly identifies the functions of the labelled structures?

	Synthesising polypeptides from amino acids	Transporting proteins	Carrying out glycosylation	Secreting digestive enzymes
A	2	1	4	3
B	1	4	2	3
C	2	3	4	1
D	1	3	2	4

- 2 The diagrams show short sections of some common polysaccharides and modified polysaccharides.



The polysaccharides can be described as below.

- Polysaccharide **F** is composed of β -glucose monomers with 1,4 glycosidic bonds.
- Polysaccharide **G** is composed of α -glucose monomers with 1,4 and 1,6 glycosidic bonds.
- Polysaccharide **H** is composed of N-acetylglucosamine and N-acetylmuramic acid monomers with β -1,4 glycosidic bonds.
- Polysaccharide **J** is composed of α -glucose monomers with 1,4 glycosidic bonds.
- Polysaccharide **K** is composed of N-acetylglucosamine monomers with β -1,4 glycosidic bonds.

Which shows the correct pairings of polysaccharide descriptions and diagrams?

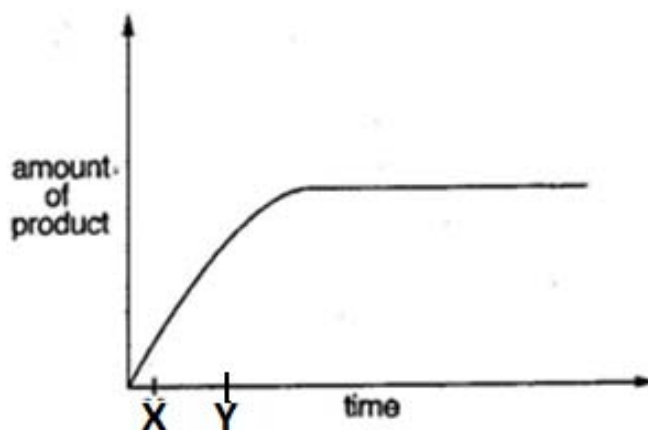
	Polysaccharide				
	F	G	H	J	K
A	2	4	5	3	1
B	2	5	4	1	3
C	3	4	1	2	5
D	3	5	4	1	2

3 With reference to carrier proteins, which of the following statements is/are true for all carrier proteins?

- 1 They contain binding sites for specific molecules or ions.
- 2 They directly require ATP to transport substances across the membrane.
- 3 They are soluble globular proteins.
- 4 They are embedded in membranes.

- A 1 only
B 1 and 4
 C 3 and 4
 D 1, 2 and 4

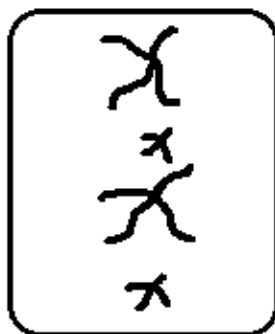
4 The graph below shows the amount of product formed in an enzyme-catalysed reaction over a certain period of time at 37° C.



What is true at time X?

- A Most enzyme molecules will have free active sites.
B The number of unreacted substrate molecules is high.
 C The number of enzyme-substrate complexes is low.
 D The rate of enzymatic reaction is lower than at time Y.

- 5 **The diagram** below shows metaphase of mitosis in a cell of an organism.



Each homologous pair of chromosomes in this organism contains 4 gene loci. This organism was genotyped and found to be heterozygous at all gene loci. The organism reproduces sexually via the production of millions of gametes by meiosis.

What is the maximum possible number of genetically different gametes that can be produced by this organism, assuming crossing over does not occur during meiosis in all cells?

- A 2
B 4
 C 16
 D 256
- 6 **Hybrid species** can be produced from cabbage and radish. The table below shows the chromosome numbers in the parental species and the hybrids.

type of cell	number of chromosomes per cell
parental cabbage	18
parental radish	18
parental gametes	9
F1 hybrids	18
F1 gametes	9
F2 hybrids	18
F2 gametes	18
F3 hybrids	36

Chromosomal mutation occurred at one stage. At which stage did it occur?

- A during the formation of the F1 gametes.
B during the formation of the F2 gametes.
 C during the fusion of the parental gametes.
 D during the fusion of the F1 gametes.

- 7 **3 different polynucleotide** molecules (X, Y and Z) were isolated from a eukaryotic cell. One of them is a double-stranded DNA gene, while the other two are the pre-mRNA and mature mRNA that the DNA gene codes for.

The adenine nucleotide content of all 3 molecules was examined and shown in the table below:

Molecule	Percentage of adenine nucleotides in the molecule / %
X	49
Y	52
Z	53

Based on the information given, which of the following conclusions is/are valid and true?

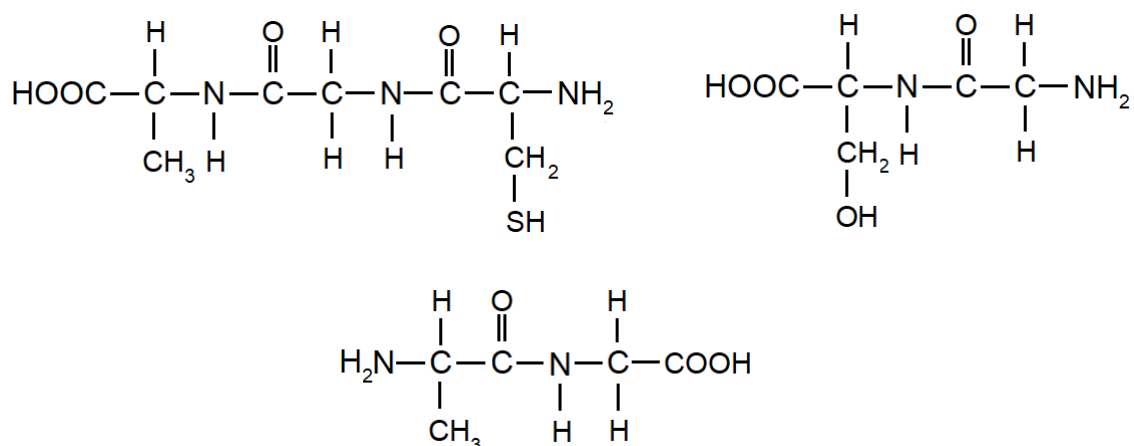
- 1 X is definitely the DNA gene.
 - 2 Z is definitely the mature mRNA.
 - 3 The pre-mRNA molecule has more uracil than guanine in it.
 - 4 Y has more purine nucleotides than pyrimidine nucleotides in it.
- A** 1 and 2
B 1, 2 and 4
C 1, 3 and 4
D 2, 3 and 4

- 8 **In an experiment**, polypeptide A, which is coded for by a non-mutated version of a prokaryotic gene, was cleaved by a particular protease.

The cleavage produced 2 fragments, one of which contains the C-terminus of polypeptide A and is 5 amino acids long. This 5-amino-acid-long fragment, now called Peptide B, was then isolated for further investigation.

A solution containing many molecules of Peptide B was treated with another protease, called Protease X. The solution was analysed after the treatment and was found to contain various different fragments of different lengths and sequences.

The structures of 3 of these fragments are shown below:



It is known that Protease X is able to cleave any peptide bonds within the molecule of Peptide B. However, the cleavage of all peptide bonds within a single molecule is rare.

The mRNA codons involved in the synthesis of the Peptide B portion of Polypeptide A are shown below:

Amino acid	R group	mRNA codon
Glycine	H	5' – GGC – 3'
Alanine	CH ₃	5' – GCC – 3'
Serine	CH ₂ OH	5' – UCC – 3'
Cysteine	CH ₂ SH	5' – UGU – 3'

Which of the following correctly shows a single point mutation in the portion of the template DNA sequence that codes for Peptide B, leading to a single amino acid substitution?

- A** 5' – GCC GCC ACA GGA GCC – 3'
B 5' – GCC GGA ACA GCC GCC – 3'
C 5' – GGA GCC GCC GCC ACA – 3'
D 5' – ACA GCC GCC GCC GGA – 3'

Examiner's comments:

C – Ala – Gly – Cys – N

C – Ser – Gly – N

N – Ala – Gly – C

Peptide B: N – Cys – Gly – Ala – Gly – Ser – C

mRNA sequence: 5' UGU GGC GCC GGC UCC 3'

Template DNA: 3' ACA CCG CGG CCG AGG 5'

Template DNA: 5' GGA GCC GGC GCC ACA 3'

- 9 Arrange the following statements on the signal transduction pathway for insulin in order.

- 1 Auto-crossphosphorylation
- 2 Increase in uptake of glucose through facilitated diffusion
- 3 Relay proteins bind to specific activated tyrosine residues
- 4 Activated relay proteins activate their respective transduction pathways
- 5 Insulin binds to receptor tyrosine kinase (RTK) at the receptor site
- 6 Vesicles containing glucose transporters move to and fuse with the plasma membrane
- 7 Changes in the 3D conformation activates the tyrosine kinase domain of receptor

- A 5, 1, 7, 3, 4, 6, 2
 B 5, 7, 1, 3, 4, 6, 2
 C 2, 5, 1, 7, 3, 4, 6
 D 2, 5, 1, 7, 4, 3, 6

- 10 The Southern pine beetle is a pest native to pine forests in Central America and the southeastern U.S..

However recent observations show the latitude of this pest infestation creeping northward by about 40 miles a decade since 1980, and could damage 273,000 square miles of pine forests by 2080.

Which of the following explanations for the above observation are attributed to climate change?

- 1 Longer and more intense droughts weakening the defenses of trees, making them vulnerable to attack by the beetles.
- 2 Long-term suppression of forest fires leaving pine forests unnaturally dense and uniform, facilitating the beetles' spread from tree to tree.
- 3 Pines trees colonising new territories with cooler climates.
- 4 Increased temperatures in the winter allowing the beetles' larvae to survive.

- A 1 and 4 only
 B 2 and 3 only
 C 1, 3 and 4 only
 D All of the above

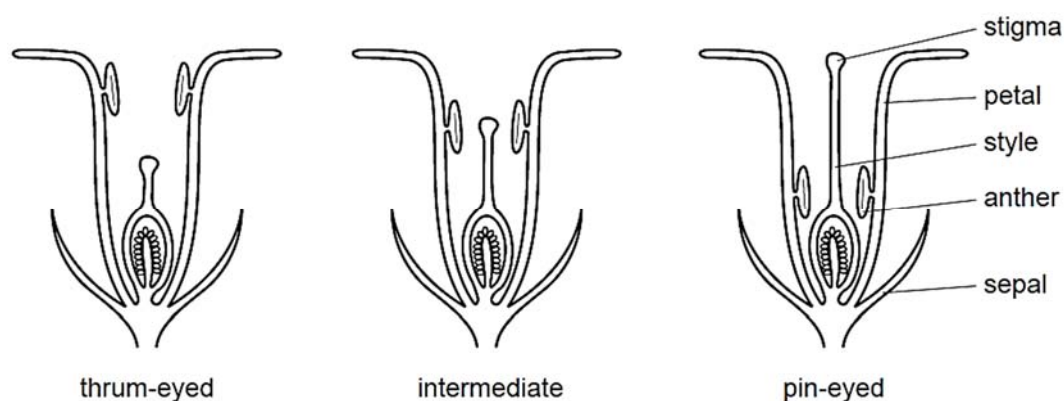
- 11 The primrose, *Primula vulgaris*, is a small herbaceous, yellow-flowered plant which is common in cooler areas of the Northern hemisphere including alpine and Arctic areas.

The flowers of the primrose have different flower shapes (polymorphic), which are adaptations for pollination. 'Thrum-eyed' primroses have a short style. 'Pin-eyed' primroses have much longer styles. The anther position also varies among the primrose.

Some populations of primrose consist almost entirely of plants with intermediate flowers. These populations are common where there are fewer winged insects.

Anthers produce pollen (male gametes) which land on the stigma, leading to fertilization.

The diagrams show polymorphic flowers of primroses.



Which statements are correct?

- 1 Cross-pollination will be favoured between pin-eyed and thrum-eyed primroses.
- 2 Primroses with pin-eyed flowers are likely to show more genetic diversity than primroses with intermediate flowers.
- 3 Primroses with thrum-eyed flowers are likely to be more able to adapt to changing environmental conditions than pin-eyed primroses.
- 4 Self-pollination is more likely to occur in primroses with intermediate flowers.

- A** 1 and 2
B 3 and 4
C 1, 2 and 4
D All of the above

- 12 On the tiny Lord Howe Island, 600 miles east of Australia, there are two species of palm which seem, from DNA analysis, to be descended from one original species. Factors involved in this speciation on this tiny island include:

- 1 linkage of genes for soil tolerance and flowering time
- 2 variation in flowering time
- 3 variation of soil tolerance
- 4 variation of soil types on the island

What is the correct sequence to explain this speciation?

- A 1 → 2 → 3 → 4
- B 2 → 1 → 4 → 3
- C 3 → 4 → 1 → 2
- D 4 → 3 → 2 → 1

Examiner's comments:

Statement 1: genes are inherited together, because plants will not be able to exchange alleles with the other variants.

- 13 Many types of evidence, can provide support for Darwin's theory of natural selection and descent with modification.

What statement provides support?

- 1 The allele for sickle cell haemoglobin that gives resistance to malaria is more frequent in malarial areas.
- 2 The distribution of the variants of the A blood group antigen reflects human migration patterns.
- 3 The homozygous condition of the sex-linked allele for a non-functional blood clotting protein is rare.
- 4 The molecular structure of ATP is almost identical in all eukaryotes.

- A 4 only
- B 2 and 3
- C 1, 3 and 4
- D 1, 2, 3 and 4

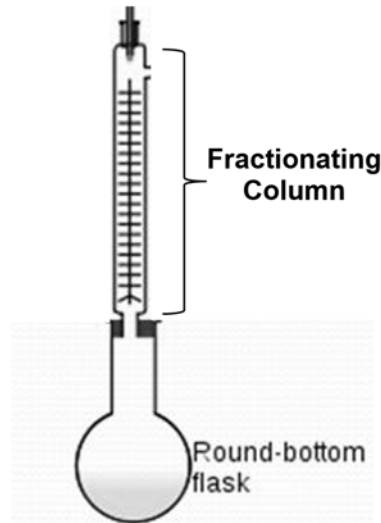
- 14 Which of the following statement(s) about chromosome structure is/are true?

- 1 Euchromatin is the more diffuse region of the interphase chromatin and is transcriptionally active.
- 2 Nucleosomes and linker DNA make up a 30nm chromatin fibre.
- 3 There are 8 nucleosomes in one turn of the helix of the 30nm chromatin fibre.
- 4 Further condensation of the 300nm chromatin fibre only takes place during mitosis/meiosis.

- A 1 and 2
- B 1 and 4
- C 2 and 3
- D 3 and 4

- 15 **The active** messenger RNAs (active mRNAs) in tissue cells can be isolated by passing the homogenised cell contents through a fractionating column (shown in diagram below). The column has short lengths of uracil nucleotides attached to a solid supporting material.

Most molecules of mRNA that pass through the column break up into small pieces and cannot be translated.



The active mRNAs that attach to the column can be separated by appropriate treatment.

Which statements correctly describe active mRNA?

- 1 Active mRNAs are held to the fractionating column by bonds between adenine and uracil bases.
- 2 Active mRNAs can be released from the fractionating column by breaking hydrogen bonds.
- 3 Only mRNAs with polyadenine tailing can be translated.

- A** 1 only
B 1 and 2
C 2 and 3
D 1, 2 and 3

16 Four different genes are regulated in different ways.

Gene 1 undergoes tissue-specific patterns of alternative splicing.

Gene 2 is part of a group of structural genes controlled by the same regulatory sequences.

Gene 3 is in some circumstances subject to methylation.

Gene 4 codes for a repressor protein which acts at an operator site close by.

Which role of the table correctly identifies which genes are prokaryotic and which are eukaryotic?

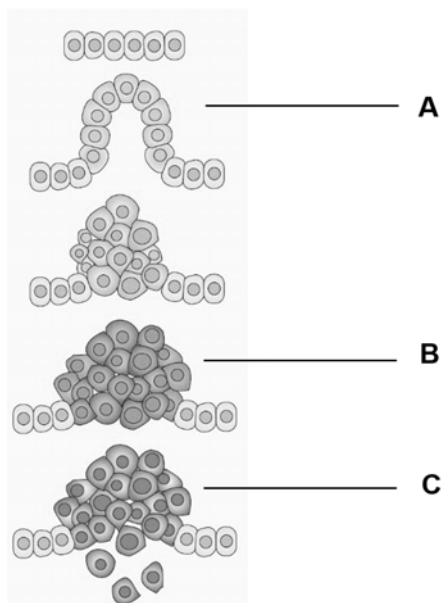
	prokaryotic	eukaryotic
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 4	1 and 3
D	2 and 3	1 and 4

17 Which of the following statements about the eukaryotic control elements are correct?

- 1 Attachment of the RNA polymerase II at the TATA box is achieved with the help of a series of specific transcription factors
- 2 A given gene may have multiple enhancers, each active at a different time or in a different cell type or location in the organism
- 3 Repressors bind to silencer regions of DNA far upstream of promoters to repress transcription

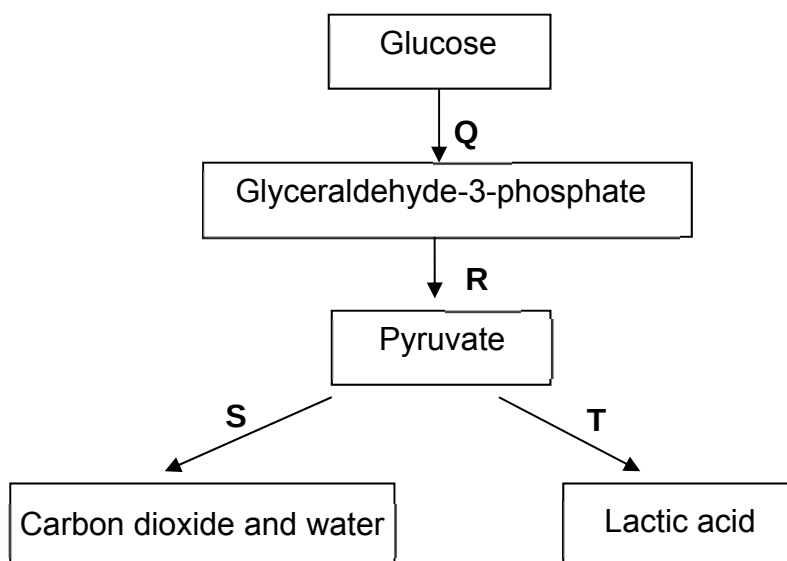
- A 1 and 3
 B 1 and 2
 C 2 and 3
 D 1, 2 and 3

- 18** The diagram below shows the multi-step model of cancer development in colon cancer. Which of the following contains the most appropriate explanation for the different stages?



	A	B	C
A	Mutation in one copy of a tumor suppressor gene	Mutation in other genes such as telomerase gene	Loss of anchorage dependence
B	Mutation in one copy of a proto-oncogene	Loss of density dependence	Loss of anchorage dependence
C	Mutation in one copy of a proto-oncogene	Loss of anchorage dependence	Loss of density dependence
D	Mutation in promoter region upstream of a proto-oncogene	Mutation in one copy of a tumor suppressor gene	Loss of density dependence

- 19 **With reference** to the diagram below, relate processes Q, R, S, T to statements (1), (2) and (3).



- (1) NAD is regenerated without the use of the electron transport system
 (2) ATP is synthesised via substrate level phosphorylation
 (3) It can take place under anaerobic conditions.

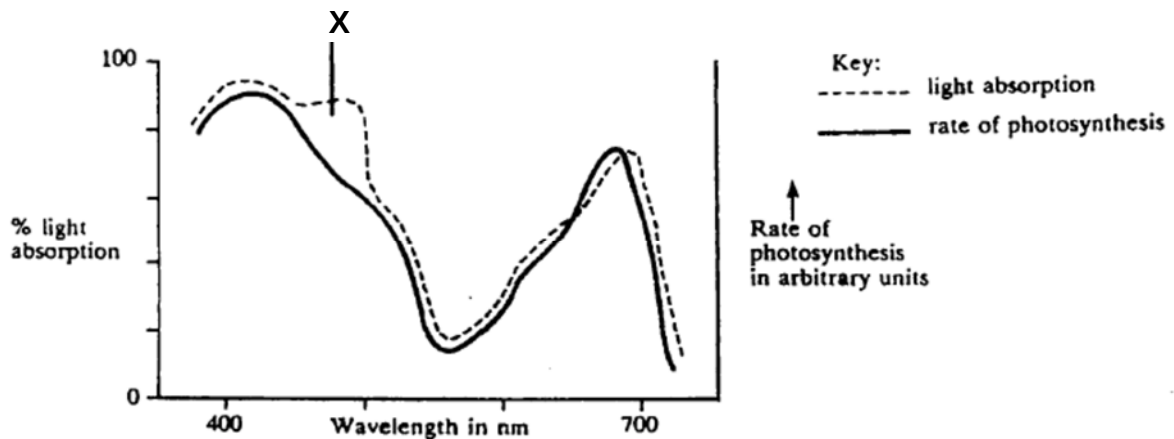
	(1)	(2)	(3)
A	T only	R only	Q,R,T only
B	T only	R,S only	Q,R,T only
C	S,T only	R only	Q,R,S,T
D	S,T only	R,S only	Q,R,S,T

A: to catch students who forgot that Sustrate-Level Phosphorylation also occurs in S (link reaction, Krebs cycle, OP)

D: to catch students who thought S is fermentation of yeast/plants

C: to catch students who thought that S is fermentation AND also forgot SLP also occur in S

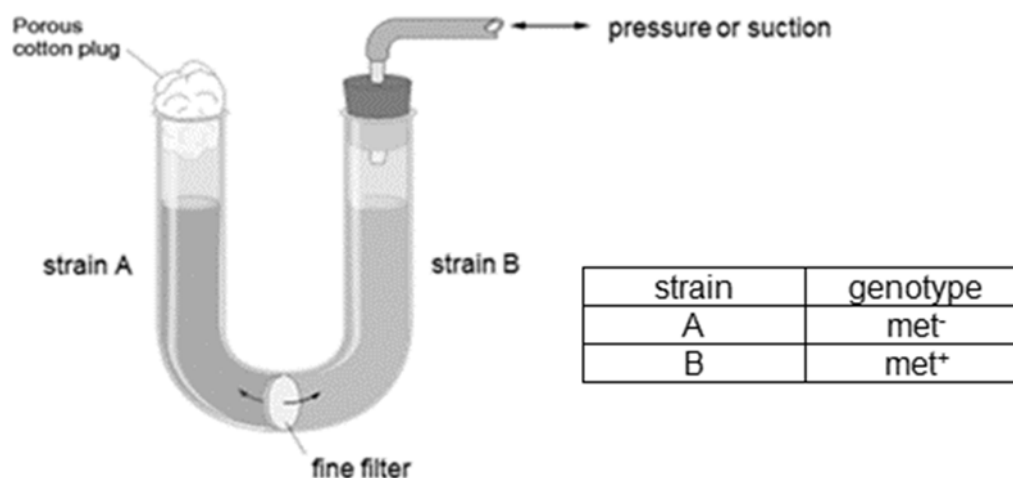
- 20 The graph below shows the effect of different wavelengths of light on the rate of photosynthesis and on the amount of light absorbed by the pigments in a green seaweed.



The difference between the two curves at X is due to

- A inefficient trapping of light energy by the chlorophyll
 - B no ATP production at that wavelength
 - C oxygen given off during photosynthesis interferes with the absorption of light.
 - D carotenes absorbing light that is not used in photosynthesis.
- 21 A mutation that renders the product of a regulatory gene non-functional for an inducible operon will result in
- A continuous transcription of the genes of the operon.
 - B irreversible binding of the repressor to the operon.
 - C complete blocking of the attachment of RNA polymerase to the promoter.
 - D continuous production of gene products that are anabolic in function

- 22 **To investigate** gene transfer between bacteria, two strains of the same bacterial species were each placed in one arm of a U-tube with a filter separating them.



met⁺ is a wild-type gene that codes for the ability to synthesise the essential amino acid, methionine.

met⁻ indicates that the met⁺ gene has been mutated.

Liquid may be transferred between the arms of the tube by the application of pressure or suction, but particles that are larger than the filter pore size would not be able to pass through the fine filter.

type of particle	size
bacteria	1 – 10µm
bacteriophages	0.025 – 0.2µm

After several hours of incubation, bacterial cells from the left arm of the tube are plated on minimal medium.

Which pair of experimental results best shows that transduction was most likely the process responsible for gene transfer between strains A and B?

	filter pore size	growth of colonies on minimal medium
A	5µm	no
	0.1µm	yes
B	5µm	no
	0.1µm	no
C	0.45µm	no
	0.02µm	yes
D	0.45µm	yes
	0.02µm	no

- 23 **Probes** are short, single-stranded DNA segments that are used to identify DNA fragments with a particular sequence. Which of the following statements about probes is false?
- A They have the same sequence as the sequence to be identified.
 - B Probes may not adhere 100% to target sequences.
 - C A probe from one organism may be used to locate a homologous DNA segment from a different organism.
 - D In order to be useful, the probe must be labelled.
- 24 **Which is** a correct statement about obtaining human embryonic stem cells for research?
- 1 Removal of these cells is considered to be ethically acceptable as normal development of the embryo is not inhibited.
 - 2 The cells must be removed at an early stage of development from a region of the blastocyst known as the inner cell mass.
 - 3 The cells must be removed immediately following the successful fertilisation of the ovum by the sperm, and after checking for normal mitotic division.
 - 4 The region of the blastocyst from where the cells are removed is an area that develops at a later stage into the placenta.
- A 2 only
 - B 1 and 2
 - C 2 and 3
 - D 3 and 4
- 25 How do viruses cause diseases in animals?
- 1 They inhibit normal synthesis of host cell DNA, RNA, or protein.
 - 2 They degrade the host cell's chromosomes.
 - 3 They disrupt the oncogenes of the host cell causing uncontrolled cell division.
 - 4 Their viral proteins and glycoproteins on the surface membrane of host cells cause them to be recognized and destroyed by the body's immune defences.
- A 1 and 3
 - B 2 and 3
 - C 1, 2 and 4
 - D 1, 2, 3 and 4

26 Which of the following statement(s) is/are true for tuberculosis?

- 1 The pathogen is from the genus *Orthopoxvirus*
- 2 Some individuals with the disease are not infectious
- 3 The disease can exist in the active or inactive state
- 4 Transmission of the disease increases with frequency of exposure to an infectious individual

- A 4 only
 B 1 and 3 only
 C 2, 3 and 4 only
 D All of the above

27 The length of the petiole (leaf stalk) in a type of flowering plant is controlled by two genes, A and B. These genes are found on different loci on non-homologous chromosomes.

Homozygous dominant plants have long petioles (30 cm), homozygous recessive plants have short petioles (10 cm). Each dominant allele contributes 5cm to the petiole length.

F₁ plants with medium length petioles (20 cm) were obtained when a plant with short petiole is crossed with a plant with long petiole. If the F₁ generation plants were allowed to cross, what proportion of their offspring would be expected to have medium length (20 cm) petioles?

- A 0.0625
 B 0.25
 C 0.375
 D 0.5

Explanation: $AaBb \times AaBb \rightarrow$ Out of 16 possibilities, look for genotype with any 2 dominant alleles and any 2 recessive alleles. $6/16 = 0.375$

28 Agouti mice have banded hairs, giving a grey colour. Black mice have unbanded hairs. White mice have no pigment. A cross between a homozygous black mouse and a white mouse produced offspring with agouti hair. Another cross between the (same) black mouse and another white mouse produced some offspring with agouti hair and some with black hair.

What explains these observations on the phenotype of hair of mice?

- A There is a single gene with two codominant alleles, black and white.
 B There is a single gene with three alleles in a dominance series, black $\rightarrow$ grey $\rightarrow$ white.
 C There are two epistatic genes, one controlling pigment production and one controlling banding.
 D There are two linked genes, one controlling pigment production and one controlling banding.

Explanation:

Scenario A: Cross 1 and 2: AA x aa → Aa (all gray)

Scenario B: Cross 1 and 2: A^BA^B x A^WA^W → A^BA^W (agouti);

Scenario C: A_ pigment production; B_ banding: AAbb x aaBB → AaBb all agouti;

AAbb x aaBb → AaBb, Aabb agouti and black

Scenario D: Linkage + epistasis?

- 29 **The Himalayan rabbits** have white hair on the body and black hair on the extremities such as feet, tail, ears and face.

The allele for the Himalayan rabbit pigment pattern, c^h, is recessive to the alleles for normal colour (all hair agouti), C, as well as dark chinchilla (all hair dark grey), c^{chd}, and is dominant to the allele for albino (all hair white, no pigment production), c. All of the alleles of this gene produce different versions of the same enzyme involved in pigment production.

A patch of white fur was removed from a Himalayan rabbit and an ice pack secured to the skin. The fur that grew back on the patch was black.

Which is correct?

	Genotypes of Himalayan rabbits	Explanation for pigment pattern in Himalayan rabbits
A	c ^h c ^h only	The enzyme is denatured at the high skin temperatures found on the rabbit's bodies
B	c ^h c ^h only	The enzyme becomes inactive at the low skin temperatures found on the rabbit's feet, tail, ears and face.
C	c ^h c ^h and c ^h c only	The enzyme is denatured at the high skin temperatures found on the rabbit's bodies
D	c ^h c ^h and c ^h c only	The enzyme becomes inactive at the low skin temperatures found on the rabbit's feet, tail, ears and face.

- 30 **Table 1 below** shows the effect of a drug called P8 on the blood pressure of mice, 20 minutes after P8 was fed to the mice. 4 mice were administered P8, and another 4 mice were given placebo drug as control.

Table 1: Effect of P8 on the blood pressure of mice

	Systolic blood pressure /mmHg				
Condition	Reading 1	Reading 2	Reading 3	Reading 4	Average
Control	175	175.5	175	176	175.5
P8 added	140	140	141	140	140.25

The mathematical formulae for standard deviation and experimental t-value are provided below:

standard deviation $s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$

Legend

Σ is summation of

x is observed values

$\bar{x}$ is the mean

n is the sample size (number of observations per condition)

$$t = \frac{(X_1 - X_2)}{\sqrt{\frac{(s_1)^2}{n_1} + \frac{(s_2)^2}{n_2}}}$$

Where:

- x1 is the mean of sample 1
- s1 is the standard deviation of sample 1
- n1 is the sample size of sample 1
- x2 is the mean of sample 2
- s2 is the standard deviation of sample 2
- n2 is the sample size in sample 2

The table below shows the Student's t-test table of t critical values:

df	.10	.05
1	3.078	6.314
2	1.886	2.920
3	1.638	2.353
4	1.533	2.132
5	1.476	2.015
6	1.440	1.943
7	1.415	1.895
8	1.397	1.860
9	1.383	1.833
10	1.372	1.812
11	1.363	1.796
12	1.356	1.782
13	1.350	1.771
14	1.345	1.761
15	1.341	1.753

Which of the following is false?

- A Degree of freedom is 1
- B Standard deviation for control group is 0.577
- C t-experimental value is 81.465
- D There is significant difference between control mice and mice administered P8

END OF PAPER